

HKUST MATH 2431 - problem set 7

3.

3. (*) Chernoff bounds

[4 Points]

Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space and $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ a random variable such that the moment generating function $\psi_X(t)$, defined by

$$\psi_X(t) = \mathbf{E}[e^{tX}],$$

exists for every $t \in \mathbb{R}$.

(a) Let $a \in \mathbb{R}$. Show that for $t > 0$, we have

$$\mathbf{P}[X \geq a] \leq e^{-ta} \psi_X(t),$$

whereas for $t < 0$, we have

$$\mathbf{P}[X \leq a] \leq e^{-ta} \psi_X(t).$$

Hint: The function $\exp : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing. Use Markov's inequality.

3.a. Note $e^{tx} > 0$.
Apply Markov's inequality: $\mathbf{P}[e^{tx} \geq e^{ta}] \leq \frac{\mathbf{E}[e^{tx}]}{e^{ta}} = e^{-ta} \psi_X(t)$
For $t > 0$, $(e^{tx} \geq e^{ta} \Leftrightarrow x \geq a) \Rightarrow \mathbf{P}[X \geq a] \leq e^{-ta} \psi_X(t)$
 $t < 0$, $(e^{tx} \geq e^{ta} \Leftrightarrow x \leq a) \Rightarrow \mathbf{P}[X \leq a] \leq e^{-ta} \psi_X(t)$

(b) Calculate $\psi_X(t)$ for $X \sim \mathcal{N}(0, 1)$.

$$\begin{aligned} 3.b. \quad \mathbf{P}[e^{tx}] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} e^{-\frac{x^2}{2}} dx & e^{tx} e^{-\frac{x^2}{2}} \\ &= e^{\frac{t^2}{2}} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{(x-t)^2}{2}} dx & = e^{-\frac{x^2-2tx}{2}} \\ &= e^{\frac{t^2}{2}} \underbrace{\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{(x-t)^2}{2}} dx}_{\text{pdf of } \mathcal{N}(t, 1)} & = e^{\frac{(x-t)^2-t^2}{2}} \\ & & = e^{\frac{t^2}{2}} e^{-\frac{(x-t)^2}{2}} \end{aligned}$$

(c) Show that for $X \sim \mathcal{N}(0, 1)$, we have

$$\mathbf{P}[X \geq a] \leq e^{-\frac{a^2}{2}}, \text{ for } a > 0,$$

and

$$\mathbf{P}[X \leq a] \leq e^{-\frac{a^2}{2}}, \text{ for } a < 0.$$

$$\text{3.c. By (a), } \begin{aligned} \mathbf{P}[X \geq a] &\leq e^{-ta} \psi_X(t) & t > 0 \\ \mathbf{P}[X \leq a] &\leq e^{-ta} \psi_X(t) & t < 0 \end{aligned} \quad \psi_X(t) = e^{-\frac{t^2}{2}}$$

sub $t = a$.

$$\text{for } a = t > 0, \quad \mathbf{P}[X \geq a] \leq e^{-a^2} \psi_X(a) = e^{-\frac{a^2}{2}}$$

$$\text{for } a = t < 0, \quad \mathbf{P}[X \leq a] \leq e^{-a^2} \psi_X(a) = e^{-\frac{a^2}{2}}$$

4.

4. (*) Joint distributions

[4 Points]

- (a) For the examples i. and ii. below, decide (with a mathematical explanation) whether the function $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ can be the joint cumulative distribution function of a pair (X, Y) of random variables. For the case where F is indeed a joint cumulative distribution function, determine the cumulative distribution functions of X and Y .

$$\text{i. } F(x, y) = \begin{cases} 1 - e^{-x-y}, & x, y \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

4.a.i. 1. $\lim_{x,y \rightarrow -\infty} F(x,y) = 0$ (consider $x < 0$ or $y < 0$, or both)

2. $\lim_{x,y \rightarrow +\infty} F(x,y) = \lim_{x,y \rightarrow +\infty} (1 - e^{-x-y}) = 1 - \lim_{z \rightarrow +\infty} e^{-z} = 1 - 0 = 1$

3. $\lim_{\substack{\epsilon_1, \epsilon_2 > 0 \\ (\text{jointly})}} F(x+\epsilon_1, x+\epsilon_2) = \lim_{\substack{\epsilon_1, \epsilon_2 > 0 \\ (\text{jointly})}} (1 - e^{-(x+y) - (\epsilon_1 + \epsilon_2)})$
 $= 1 - \lim_{z > 0} e^{-(x+y) - z} = 1 - e^{-(x+y)} = F(x,y)$

4. For any $\epsilon_1, \epsilon_2 > 0$

$$\begin{aligned} & F(x+\epsilon_1, y+\epsilon_2) - F(x+\epsilon_1, y) - F(x, y+\epsilon_2) + F(x, y) \\ &= -e^{-(x+y)} (e^{-(\epsilon_1 + \epsilon_2)} - e^{-\epsilon_1} - e^{-\epsilon_2} + 1) \\ &= -e^{-(x+y + \epsilon_1 + \epsilon_2)} (1 - e^{\epsilon_2} - e^{\epsilon_1} + e^{\epsilon_1 + \epsilon_2}) \\ &\quad \underbrace{> 0} \quad \underbrace{(e^{\epsilon_1} - 1)(e^{\epsilon_2} - 1)}_{> 0} > 0 \\ &< 0 \text{ (problematic)} > 0 \quad > 0 \end{aligned}$$

So $F(x,y)$ is NOT a CDF

ii.
$$F(x, y) = \begin{cases} 1 - e^{-x} - xe^{-y}, & 0 \leq x \leq y \\ 1 - e^{-y} - ye^{-x}, & 0 \leq y \leq x \\ 0, & \text{otherwise.} \end{cases}$$

4. a. ii. 1. $\lim_{x,y \rightarrow -\infty} F(x,y) = 0$ (consider $x < 0$, $y < 0$, or both)

2. $\lim_{x,y \rightarrow +\infty} F(x,y) = 1$

check first two branches:

$$\text{for } 0 \leq x \leq y \quad \lim_{x,y \rightarrow +\infty} 1 - \underbrace{e^{-x}}_{\rightarrow 0} - \underbrace{xe^{-y}}_{\rightarrow 0} = 1$$

$$xe^{-x} \geq xe^{-y} \quad \text{as } x \rightarrow +\infty$$

$$xe^{-x} \rightarrow 0, \text{ so } xe^{-y} \rightarrow 0$$

$$\text{for } 0 \leq y \leq x \quad \lim_{x,y \rightarrow +\infty} 1 - \underbrace{e^{-y}}_{\rightarrow 0} - \underbrace{ye^{-x}}_{\rightarrow 0} = 1$$

3. exp is cont.

cont. is preserved under $+$, $-$, $\cdot$ and function composition

So each of the three branches is cont.

At the boundaries: $x=0$, $F(x,y)=0$ for all applicable branches

$y=0$, $F(x,y)=0$ for all applicable branches

(in particular, $F(0,0)=0$)

$x=y \geq 0$, the ^{first} two branches are the same expressions

So $F(x,y)$ is cont., in particular right-continuous jointly.

$$4. \quad \nabla F(x,y) = \begin{cases} (e^{-x} - e^{-y}, xe^{-y}) & 0 < x \leq y \\ (0, ye^{-x}) & 0 < y \leq x \\ (0,0) & x < 0 \text{ or } y < 0 \\ \text{undefined} & \text{otherwise} \end{cases} \quad \left. \begin{array}{l} \text{valid when } x=y, \\ \text{evolves to the} \\ \text{same vector} \end{array} \right\}$$

$$\text{Now, obviously } F(x+\epsilon_1, y+\epsilon_2) \geq F(x, y+\epsilon_2) + F(x+\epsilon_1, y) - F(x,y) \\ \text{for } x \leq 0 \text{ or } y \leq 0$$

Check $x, y > 0$, check the partial mixed derivatives.

$$\text{for } 0 < x \leq y, \quad \partial_x \partial_y F = \partial_y \partial_x F = e^{-y} > 0$$

$$\text{for } 0 < y \leq x, \quad \partial_x \partial_y F = \partial_y \partial_x F = 0$$

$$\text{So } \partial_x \partial_y F = \partial_y \partial_x F \geq 0$$

so it respects the rect. inequality.

So F is a valid joint CDF.

for $x > 0$

$$F_X(x) = \lim_{y \rightarrow +\infty} F(x, y) = \lim_{y \rightarrow +\infty} 1 - e^{-x} - \underbrace{xe^{-y}}_{\rightarrow 0} \quad y \geq x \text{ eventually}$$

For $y > 0$

$$= 1 - e^{-x}$$

$$F_Y(y) = \lim_{x \rightarrow +\infty} F(x, y) = \lim_{x \rightarrow +\infty} 1 - e^{-y} - \underbrace{ye^{-x}}_{\rightarrow 0} \quad x \geq y \text{ eventually}$$

$$= 1 - e^{-y}$$

Similarly, for $x \leq 0$, $F_X(x) = 0$
 $y \leq 0$, $F_Y(y) = 0$

(b) Let X, Y be discrete random variables with joint probability mass function

$$p_{X,Y}(j, k) = \mathbf{P}[X = j, Y = k] = \begin{cases} C \cdot \left(\frac{1}{2}\right)^k, & \text{for } k = 2, 3, \dots, j = 1, 2, \dots, k-1, \\ 0, & \text{otherwise.} \end{cases}$$

Determine the constant C and the marginal probability mass functions p_X and p_Y .

4.b. require $1 = \sum_{k=2}^{+\infty} \sum_{j=1}^{k-1} C \cdot \left(\frac{1}{2}\right)^k$ note for $0 < x < 1$, $\frac{1}{1-x} = 1 + x + x^2 + \dots = \sum_{k=0}^{+\infty} x^k$

$$= C \cdot \sum_{k=2}^{+\infty} (k-1) \left(\frac{1}{2}\right)^k$$

$$\frac{1}{(1-x)^2} = \left(\frac{1}{1-x}\right)' = 1 + 2x + \dots = \sum_{k=0}^{+\infty} (k+1) x^k$$

$$= C \cdot \frac{\left(\frac{1}{2}\right)^2}{\left(1 - \frac{1}{2}\right)^2} \quad (x = \frac{1}{2})$$

$$= \frac{1}{x^2} \sum_{k=2}^{+\infty} (k-1) x^k$$

$$= C \cdot \frac{1/4}{1/4}$$

$$= C$$

$$\frac{x^2}{(1-x)^2} = \sum_{k=2}^{+\infty} (k-1) x^k$$

So $C = 1$.

$$p_X(j) = \sum_{k=j+1}^{+\infty} \left(\frac{1}{2}\right)^k = \left(\frac{1}{2}\right)^j \quad (j \geq 1, 0 \text{ otherwise})$$

$$p_Y(k) = \sum_{j=1}^{k-1} \left(\frac{1}{2}\right)^k = (k-1) \left(\frac{1}{2}\right)^k \quad (k \geq 2, 0 \text{ otherwise})$$